Engineering Chemistry · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry

DRAFT — UE26CY151A · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry · Electronic spectroscopy(Progression)

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
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M1

Electronic spectroscopy(Progression)

1 Electronic spectroscopy(Progression)

Study focus: Electronic spectroscopy(Progression). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000126 · EV0000127

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module one on molecular spectroscopy and we are going to study electronic spectroscopy in this class.

Key details

- We will revisit the Born-Oppenheimer approximation.
- And then finally we will study the vibrational core structure, which is also called progression.
- In electronic spectroscopy, we will find out where the electronic levels in the molecule.
- Like in rotational spectroscopy, we talked about rotational levels.
- In vibrational spectroscopy, we talked about the vibration levels.
- In electronic spectroscopy, we will talk about the electronic levels which are there inside the molecules.
- and now when a UV visible light is made to fall on the sample, the energy is sufficient to cause electronic transitions and these transitions will happen from the highest occupied molecular orbital to the lowest unoccupied molecular orbital.
- For these electronic transitions to take place, UV visible light has the correct frequency which matches with the frequency or the energy gap between these electronic levels.
- In molecular orbital theory we studied that we have sigma molecular orbitals, pi orbitals, then we have n and then we have the pi star and then the sigma star.
- And n is known as a non-bonding orbital.
- You can also have pi to pi star and you can have n to pi star.
- If the energy gap is very high, then we it will fall in the UV region and therefore we cannot see the color.
- But if these transitions happen to fall in the visible region or the energy gap is same as that of the visible region, then we will be able to see the color.

- of the particular molecule.
- There is electron redistribution, meaning if the electron goes in one of the molecular orbitals, which is the highest molecular or highest occupied molecular orbital, and there is a transition which happens to the lowest unoccupied molecular orbital, then this there is a electron redistribution.
- which is going to change the electric field.
- And therefore we find that all molecules can undergo electronic transitions.
- If you remember rotational and vibrational spectra were not possible and we could not record those spectra because for the homonuclear diatomic molecules but.
- Electronic transitions are possible for all molecules.
- Because now we understand what do we mean by rotational levels, what we mean by vibrational levels, and what is a electronic level.
- Each of these energies can be treated independently.
- You can calculate what is the rotational energy of the molecule, what is the vibrational energy of the molecule, and what is the electronic energy of the molecule, sum them up and that will give us the total energy of the molecule.
- Now, if we are talking about change in total energy, because what we are interested in is the energy gap.

Example or application from the lecture

- Let us revisit the Born Oppenheimer approximation.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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Core explanation · 05:00–09:59

And the electronic change in energy of the molecule if this is given in joules.

Key details

- But most of the times in we express energy in spectroscopic units, therefore we write it as small E, so del E total is equal to the del E rotational, del E vibrational plus the.
- Del E electronic and remember these are now expressed in wave numbers.
- That is because the energy gap between the electronic charge electronic levels is much higher than that of the vibrational levels.
- vibration into ten to r power of three.
- So that means the energy gap in an electron in the like between electronic levels is almost thousand times that b what is there between the vibrational levels and similarly for rotational levels it is e del e rotational multiplied by ten to the power of six.
- So this is the reason why the Born Oppenheimer approximation holds because what are we what is the approximation here that each of these energies are totally independent of each other because the difference in these energy levels is very, very high.
- Within the vibrational levels we have the rotation levels.
- You can see very clearly the difference in energy is very high.
- or and within the vibration levels and rotation levels.

- That is true.
- Why because they could be in the same electronic state, but they could be in different vibrational levels within the same electronic state.
- And also they could be in different rotational levels within the same vibrational state.
- So therefore the total energy would be the sum of.
- Which electronic state it is in and what is the vibrational state and what is its rotational state.
- is saying that an electron can ha can be in the same electronic state, but the total energy could be different.
- Now electronic transition as I mentioned just now it involves electronic redistribution means the electron is now it was at one particular level and it went to another level.
- Therefore, there will be a distortion in the electrostatic forces or in distortion in the electric field which is felt by the molecule.
- It is going to vibrate.
- Whenever there is an electronic transition, there is a vibrational change also happening, and there is a rotational change also happening.
- And if we have very good resolution, we can even see the rotational changes which are happening which is referred to as the fine structure in the electronic spectr of the molecule.

Example or application from the lecture

- So therefore we can use this approximation that the total energy of the molecule can be given as a sum independent sum of the rotational vibration and electronic energies and assuming that there is no electro there is no coupling between any of these energies.

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Core explanation · 10:00–14:59

But along with the electronic transition, there could be some vibrational changes.

Key details

- If we can somehow record those vibrational changes, we can get the core structure that's called the vibrational core structure and if the resolution of our instrument is very high we can even see the fine structure.
- Remember right in the first class I said that molecular spectra are band spectrum so now you can understand that the band is the electronic transition within that when you see these lines these fine lines they are due to the core structure and further if you see finer lines then that is referred to as the.
- So that means.
- For homonuclear diatomic molecules like hydrogen and nitrogen, which were otherwise inactive in the IR and microwave region, we can record the electronic spectrum and if we can get the vibrational and the rotational structure from this electronic spectrum.
- Then we will be able to get all the parameters, the rotational parameters, the vibrational parameters for these homonuclear diatomic molecules.

- pure rotational and pure vibrational spectroscopy but unfortunately the hydrogen and nitrogen molecules which are homonuclear diatomic molecules they are not active because they don't possess dipole moment and there is no change in dipole moment when the molecule either rotates or when it vibrates.
- But all these molecules they have these molecular orbitals filled with electrons.
- So electronic transition are possible and therefore using the electronic spectrum we can get to the vibrational core structure and the fine structure and that is.
- How we can determine all the rotational and vibrational parameters for these molecules.
- How are we going to use it to determine the vibrational parameters of the molecule, though?
- We are using UV visible light, that means it is going to result in electronic transitions.
- This is the structure where we have this lower electronic state that is known as the ground state, and this will be the excited state.
- And what we notice here that the lines are becoming closer and closer.
- What are these lines representing?
- They are representing the energy levels.
- Vibrational energy levels within the electronic state.
- What does it mean that.
- These are this is representing an anharmonic oscillator and we notice that V double prime equal to zero happens to be slightly away from the e is equal to zero because we know that always molecules will possess this zero point energy.
- prime we are referring to the ground state and V prime refers to the excited electronic state.
- So, therefore, we have V double prime equal to 01234, and here we have V prime equal to 01234.
- And just like we have vibrational levels in the ground state, we will also have vibrational levels in the excited state.
- And therefore, it is very important that we distinguish in the nomenclature.
- And when I say V double prime equal to one then it is referring to the ground state.

Example or application from the lecture

- The always the ground state is given the double prime means E double prime stands for the ground electronic state and e prime always stands for the excited state.

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Core explanation · 15:00–19:59

For vibrational transitions during an electronic transition.

Key details

- If whenever there is a molecule which is going from ground state to the excited state, we must remember that we are using UV visible light, that means the transition has to happen from here to here and not within the vibrational level, say or of the same.
- electronic state no we are using UV visible light so therefore the transition has to happen to the next electronic state now it can go from zero to zero it can go from zero to one zero to

two and also it can go from V double prime equal to one and go to two three four of the higher levels.
- What does it mean?
- There are no selection rules to be followed.
- When we are talking about an electronic transition, during an electronic transition, there is no selection rule for the vibrational transition.
- But again, this fact helps us to simplify the spectrum.
- Because we know at room temperature it is V double prime equal to zero.
- Now we should not write V is equal to zero, we should say V double prime equal to zero because we are talking of two different electronic states.
- So V double prime equal to zero is the most populated state populated level and therefore all transition are going to start from v double prime equal to zero.
- you can have V double prime equal to 0 to V prime equal to 0.
- This is how the energy gap is increasing.
- All these are permitted because there is no selection rule.
- And also now all the transitions are starting from V double prime equal to zero.
- We just now saw that there will be transitions and how do we label these lines?
- It's very important to know how we are labeling them.
- comma V double prime what does it mean that the lines are going to be labeled according to the level where it goes that is this is the higher vibrational level vibration level of the high excited electronic state and this is the vibration level where it is originating.
- So therefore if there is a transition from zero to zero then we will get a line here that is zero.
- And if the transition is from zero to one.
- So it is one zero because we always call label the lines like this the higher.
- And again, we can see centimeter inverse is the scale here.
- That means the energy is going this way.
- I have to put the 00 line here first and not the 1, 0 or.
- The reverse way.
- Because the energy gap is going on increasing as we move from left to right.
- Because the numbers 0, 1, 2, 3, 4, they are changing in such a manner to where it is increasing by one for every line here.
- Understand all these lines are originating from zero.
- So therefore zero, one zero, two zero, three zero and four zero would rebuild the lines you will see in the spectrum and such a spectrum is known as a progression.

Example or application from the lecture

- Let us come to the next rule.
- Electronic state vibration level is named first, is it given first and then the lower.

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Core explanation · 20:00–24:59

fact which you see here as we go to these other hights where the energy is more we find that the lines are coming closer together.

Key details

- So that is because there is an harmonicity in the excited electronic state.
- If it was harmonic, then all these lines would be equidistant.
- But since there is an harmonicity in the upper electronic state, we find that these lines are actually converging.
- They will start coming closer and closer as we go to the higher values of energy.
- We have understood this very clearly.
- This is exactly what we drew and this is the ground state and this is the excited state.
- electronic state and there are transitions happening all our transitions are starting from v double prime equal to zero so if I say it is a line to zero that means it is starting from v double prime equal to zero and which and it is entering up at V prime equal to 2.
- So that would represent a two zero line similarly any of these lines are can be seen here these are all electronic transition and yes this requires minimum energy zero and then energy gap goes on increasing therefore the lines also will appear towards the higher energy.
- And then we have these lines coming closer together as we go to the higher values of energy, and this shows very clearly that the there is an harmonicity in the excited state.
- From this spectrum, we can get parameters.
- of the excited state very clearly.
- It appears for a very short while so therefore calculating the various parameters is going to be very difficult.
- But if you can record this electronic spectrum, then we can get all the information about the bond length, bond strength.
- If you're doing only the vibrational core structure, you can get the bond strength, and if you can do the rotational fine structure as.
- All lines for the ground state are originating from V double prime equal to zero, but nevertheless they are ending up at different levels of.
- The excited state and therefore we can get information about the excited state if we can record the spectrum.
- One is about this excited state, any transient.
- Excited state, we can get all the parameters, and for homonuclear diatomic molecules also, we can get all the information.
- In the next class, we are going to talk about the intensities of these lines.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Electronic spectroscopy(Progression) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_Progression__6dd27636f794caa1.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electronic spectroscopy(Progression) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_Progression__6dd27636f794caa1.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electronic spectroscopy(Progression) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_Progression__6dd27636f794caa1.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.
Electronic spectroscopy(Progression) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_Progression__6dd27636f794caa1.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00026	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00027	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00028	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00029	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

C Revision and retrieval practice